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CO3-AT2

1. A shepherd gets bored tending the town's flock. to have some fun he cries out, "wolf"! even though no wolf is in sight. The villagers run to protect the flock, but then get really mad when they realize the boy was playing a joke on them.
One night the shepherd boy sees a real wolf approaching the flock and call out. the town goes hungry. panic ensues identify TP, FP, FN, TN.

Act \ pred	W = Yes	W = NO
W = Y (P)	TP WOLF came ✓ Warning ✓	FN WOLF came No Warning
W = N (N)	FP WOLF NO came but waring	TN NO WOLF came ✓ NO warning ✓

Actual

Prediction

2. Assume there are 100 images, 30 of them depict a cat the rest donot. A machine learning model predicts the occurances of a cat in 25 of 30 cat Images. It also predicts absence of a cat in 50 of the 70 no cat Images. Identify TP, FP, FN, TN.

Act \ pred	Cat = positive	Nocat = Negative
cat = (positive)	TP = 25 Actual cat & predcat	FN = 5 (Actual cat but pred no cat)
Nocat (Negative)	FP = 20 (Act Nocat but predicted cat)	TN = 50 (Act Nocat & predicted Nocat)

Precision

$$\text{precision} = \frac{TP}{TP+FP}$$

$$= \frac{25}{25+20} = 0.556 \times 100 = (55.6\%)$$

Recall:

$$Recall = \frac{TP}{TP+FN}$$

$$= \frac{25}{25+5}$$

$$= 0.8333 \times 100$$

$$= 83.33\%$$

F1 score:

$$F1\ score = \frac{2 \times precision \times recall}{precision + recall}$$

$$= \frac{2 \times 0.556 \times 83.33}{0.556 + 83.33}$$

$$= 0.6667 \times 100 = 66.67\%$$

3. An e-mail service provider wants to classify emails as spam or Notspam using features

→ contains "offer" (yes/no)

→ contains "free" (yes/no)

→ email length (short/long)

Sol. 1. Total = 6

spam = 3

Not spam = 3

$$p(spam) = \frac{3}{6} = \frac{1}{2}$$

$$p(Not\ spam) = \frac{3}{6} = \frac{1}{2}$$

2. Conditional probability table

$P(O = \text{yes}/\text{spam}) = \frac{2}{3}$	$P(O = \text{yes}/\text{Nospam}) = \frac{1}{3}$
$P(O = \text{No}/\text{spam}) = \frac{1}{3}$	$P(O = \text{No}/\text{Nospam}) = \frac{2}{3}$
$P(F = \text{yes}/\text{spam}) = \frac{3}{3}$	$P(F \cdot \text{yes}/\text{Nospam}) = \frac{0}{3}$
$P(F = \text{No}/\text{spam}) = \frac{3}{0}$	$P(F \cdot \text{No}/\text{No spam}) = \frac{3}{3}$
$P(L = \text{yes}/\text{spam}) = \frac{2}{3}$	$P(L = \text{yes}/\text{Nospam}) = \frac{1}{3}$
$P(L = \text{No}/\text{spam}) = \frac{1}{3}$	$P(L = \text{No}/\text{Nospam}) = \frac{2}{3}$

3. Using Naive bayes, Classify :- $P(A/B) = \frac{P(B/A)CP(A)}{P(B)}$

Offer	Free	Length	class
yes	yes	short	spam
yes	NO	long	Notspam
NO	yes	short	spam
NO	yes	long	Not spam
yes	yes	long	spam
NO	NO	short	Notspam

for spam: p(spam) spam) × p(yes/spam) × p (p(Not spam) × p(short/spam)

$\frac{1}{2} \times \frac{2}{3} \times \frac{0}{3} \times \frac{2}{3} = 0$

For Not spam: p(Notspam) × p (Notspam) × p(yes/Notspam) × p (No/Notpam) × p (shorts/Notspam)

$\frac{1}{2} \times \frac{1}{3} \times \frac{3}{3} \times \frac{1}{3} = \frac{1}{18}.$

= 0.056 for Notspam.

$\therefore$ p(Not spam x) > p(spam x))

therefore using the navie bayes classifier the given email (offer = yes, Free = No length = short) is classified as Not spam because its probability for not spam is higher than spam

(iv) Comparing results using weka

Dataset	(1) Bayes. Ba	(2) Bayes
Iris	(100) 93.20	95.73
supermarket	((100) 63.71	63.71
	(v / / *)	(0/2/0)

key (i) Bayes. BayesNet

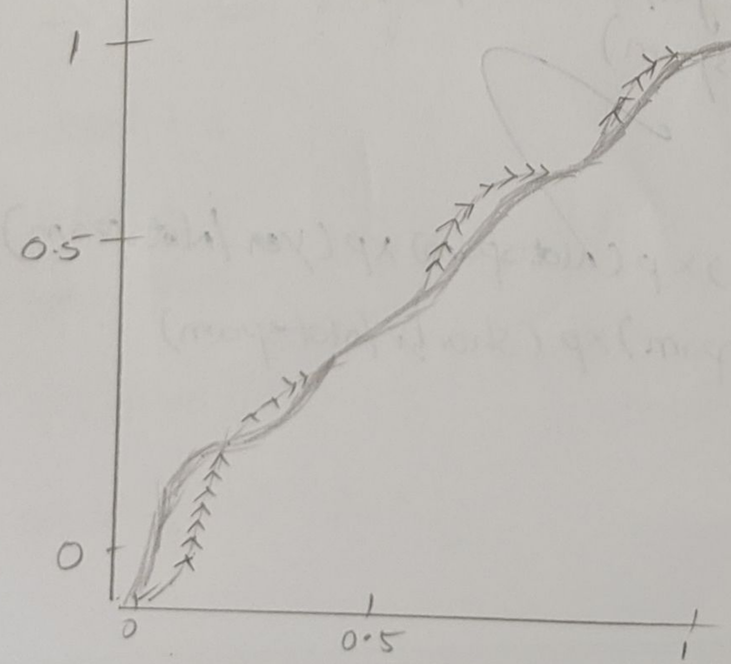
(ii) Bayes Naive bayes

Roc Curve:

X: False positive rate (Num)

Y: True positive rate (Num)

col d: thresold (Num)



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